

Supplementary appendix: Data – Committee-Based Blockchains as Games between Opportunistic Players and Adversaries

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To draw Figures 1 and 2, we collected the names and market capitalizations of the cryptocurrencies on CoinMarketCapt (<https://coinmarketcap.com/>), which is “the world’s most-referenced price-tracking website for cryptoassets.” To focus on the most significant cryptocurrencies, we only considered the 50 largest cryptocurrencies in terms of market capitalization. Among these, we counted the number of cryptocurrencies relying on committee-based blockchains. Sometimes, understanding the consensus process used by the underlying blockchain is tricky and tedious. One must go to the cryptocurrency’s blockchain source code and check which consensus protocol is used. Sometimes, external services are used, and the consensus algorithm used is not apparent. To take into account the “Merge”, the upgrade of Ethereum to proof-of-stake, we use data from the 27th of November 2022, the latest data available when collecting the data, and get the following table.

	Name	Market Capitalization in \$USD	Consensus Mechanism
1	Bitcoin	316,040,534,591.71	PoW
2	Ethereum	146,252,301,744.43	PoS – Committee-based
3	Tether	65,336,770,716.05	Multiple blockchains (Ethereum, TRON, Algorand, Avalanche, Tezos, ...) All PoS – Committee-based
4	BNB	49,179,890,751.32	PoS – Committee-based
5	USD Coin	65,336,770,716.05	Multiple blockchains (Ethereum, TRON, Algorand, Avalanche, ...) All PoS – Committee-based
6	Binance USD	22,408,551,005.70	PoS – Committee-based
7	XRP	19,959,580,740.62	PoS – Committee-based
8	Dogecoin	13,096,141,897.47	PoW

9	Cardano	10,759,011,469.76	PoS – Chain-based
10	Polygon	7,379,662,277.41	PoS – Committee-based
11	Polkadot	6,042,343,992.91	PoS – Committee-based
12	Dai	5,686,312,479.07	PoS – Committee-based (Ethereum)
13	Litecoin	5,381,161,230.15	PoW
14	Solana	5,124,488,709.71	PoS – Committee-based
15	Shiba Inu	5,091,022,513.56	PoS – Committee-based (Ethereum)
16	TRON	4,905,912,200.71	PoS – Committee-based
17	Uniswap	4,120,133,329.54	PoS – Committee-based (Ethereum)
18	UNUS SED LEO	4,016,731,620.28	Multiple blockchains (Ethereum and EOS) Both PoS – Committee-based
19	Avalanche	3,872,113,263.01	PoS – Committee-based
20	Wrapped Bitcoin	3,564,597,413.77	PoS – Committee-based (Ethereum)
21	Chainlink	3,506,270,716.55	PoS – Committee-based
22	Cosmos	2,843,328,965.41	PoS – Committee-based
23	Ethereum Classic	2,737,006,195.88	PoW
24	Monero	2,483,986,121.08	PoW
25	Stellar	2,316,510,249.13	PoS – Committee-based
26	Bitcoin Cash	2,140,406,278.93	PoW
27	Toncoin	2,113,618,445.86	PoS – Committee-based
28	Algorand	1,711,097,387.91	PoS – Committee-based
29	Cronos	1,624,231,584.13	Other / Proof-of-Authority
30	Filecoin	1,420,589,196.57	PoS – Chain-based
31	Quant	1,423,142,810.43	PoS – Committee-based (Ethereum)
32	ApeCoin	1,380,385,682.91	PoS – Committee-based (Ethereum)
33	VeChain	1,358,558,346.66	Other / Proof-of-Authority
34	NEAR Protocol	1,341,151,465.02	PoS – Committee-based
35	OKB	1,252,216,314.75	PoS – Committee-based
36	Flow	1,162,515,956.10	PoS – Committee-based
37	Hedera	1,129,191,672.31	PoS – Committee-based
38	Huobi Token	1,081,686,745.36	PoW
39	Internet Computer	1,069,605,987.38	PoS – Committee-based
40	Chiliz	1,042,505,498.53	PoS – Committee-based (Ethereum)
41	MultiversX (Elrond)	1,015,761,832.16	PoS – Committee-based
42	EOS	999,243,896.09	PoS – Committee-based
43	Terra Classic	971,774,614.53	PoS – Committee-based
44	Theta Network	954,516,542.36	PoS – Committee-based
45	Pax Dollar	947,737,831.66	PoS – Committee-based (Ethereum)
46	Tezos	909,275,455.71	PoS – Committee-based
47	Chain	881,986,380.70	PoS – Committee-based (Ethereum)
48	Aave	867,079,328.25	Multiple blockchains (Ethereum, Avalanche, ...) All PoS – Committee-based
49	The Sandbox	854,529,557.88	PoS – Committee-based (Ethereum)
50	Trust Wallet Token	828,110,812.92	PoS – Committee-based (Binance)

Using the above table, we computed the following different values:

- Market capitalization of cryptocurrencies using a PoW blockchain: \$342,960,923,060.58.
- Market capitalization of cryptocurrencies using a PoS blockchain: \$440,879,668,442.95.
- Market capitalization of cryptocurrencies using neither PoS nor PoW: \$2,982,789,930.79.

Using the above values, we drew Figure 1. Notice that only PoW and PoS are represented in the figure. Indeed, the Others category represents only 0.38% of the market capitalization in the Top 50, therefore, for clarity, it does not appear on the figure. We noticed while conducting this analysis that, except for Cronos and VeChain, all cryptocurrencies relying on committee-based blockchains were based on PoS. We have the following data:

- Market capitalization of cryptocurrencies in the Top 50 relying on chain-based PoS: \$14,496,110,915.46.

- Market capitalization of of cryptocurrencies in the Top 50 relying on committee-based PoS: \$426,383,557,527.49.

We used these values to draw Figure 2.